

MEF Assignment 3 Group42

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Question 1

Using the sample mean vector $\hat{\mu}$ and covariance matrix $\hat{\Sigma}$ of the 25 portfolio returns, we solved the quadratic optimization problem

$$\min_w \frac{\lambda}{2} w^\top \hat{\Sigma} w - \hat{\mu}^\top w \quad s.t. \quad w^\top \mathbf{1} = 1$$

for 250 values of $\log \lambda$ equally spaced on $[0.01, 4]$. For each value of λ , We computed the corresponding portfolio mean $\hat{\mu}_P$ and standard deviation $\hat{\sigma}_P$ and plotted the resulting expected-utility frontier in the $(\hat{\sigma}_P, \hat{\mu}_P)$ -plane.

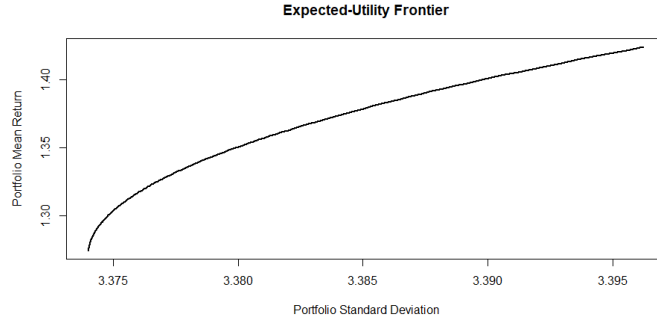


Figure 1: Expected-utility frontier in the $(\hat{\sigma}_P, \hat{\mu}_P)$ -plane.

Question 2

Next we computed the Markowitz minimum-variance set by solving

$$\min_w w^\top \hat{\Sigma} w \quad s.t. \quad w^\top \mathbf{1} = 1, \quad w^\top \hat{\mu} = \bar{\mu}$$

for 2000 target returns $\bar{\mu}$ and overlaid this frontier with the expected-utility frontier from Question 1.

The two frontiers almost coincide on the efficient part of the Markowitz set. This is consistent with the theoretical result that, under CARA utility and Gaussian returns, the expected-utility portfolio choice problem is equivalent to mean-variance optimization. Hence, varying the risk-aversion parameter λ traces out portfolios on the mean-variance frontier.

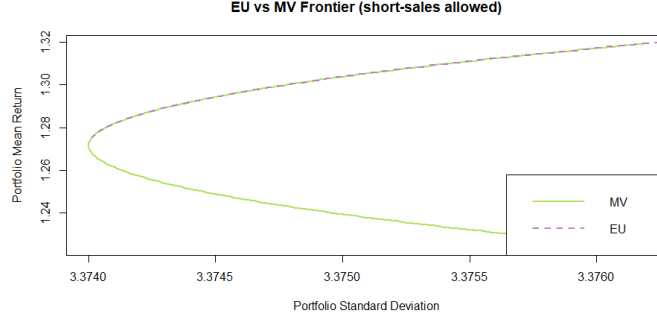


Figure 2: Overlay of the expected-utility frontier and the Markowitz minimum-variance frontier.

Question 3

For the smallest value of λ , the optimal portfolio has mean return 1.4238 and standard deviation 3.396245. For the largest value of λ , the optimal portfolio has mean return 1.2745 and standard deviation 3.374008. The global minimum-variance portfolio has mean return 1.2715 and standard deviation 3.374001.

These results show a clear pattern. When λ is small, the investor is less risk-averse and therefore chooses a portfolio with a higher expected return at the cost of a higher standard deviation. When λ is large, risk is penalized more heavily, so the optimal portfolio moves very close to the global minimum-variance portfolio. As $\lambda \rightarrow \infty$ the solution of the expected-utility problem converges to the global minimum-variance portfolio.

Portfolio	λ	Mean	Standard Deviation
EU (smallest λ)	1.01	1.4238	3.396245
EU (largest λ)	54.60	1.2745	3.374008
GMV	—	1.2715	3.374001

Table 1: Comparison of optimal portfolios for different levels of risk aversion

Question 4

We estimated the two linear SDF specifications by GMM using the gross returns of portfolios P06–P25 as test assets.

For the CCAPM specification,

$$m_t = \theta_0 + \theta_1 G_t^c,$$

the estimates are $\hat{\theta}_0 = 0.1475$ and $\hat{\theta}_1 = 0.8400$, with standard errors 22.3839 and 22.3325, respectively. The J-statistic is 25.938 with p-value 0.1012.

For the FF3 specification,

$$m_t = \theta_0 + \theta_1 RM_t + \theta_2 SMB_t + \theta_3 HML_t,$$

the estimates are $\hat{\theta}_0 = 0.3079$, $\hat{\theta}_1 = 0.3939$, $\hat{\theta}_2 = 0.1375$, and $\hat{\theta}_3 = 0.1461$, with standard errors 2.8898, 2.1005, 1.4761, and 1.6623, respectively. The J-statistic is 26.425 with p-value 0.0483.

Model	Parameter	Estimate	Std. Error	J-stat	p-value
CCAPM	θ_0	0.1475	22.3839	25.938	0.1012
CCAPM	θ_1	0.8400	22.3325	25.938	0.1012
FF3	θ_0	0.3079	2.8898	26.425	0.0483
FF3	θ_1	0.3939	2.1005	26.425	0.0483
FF3	θ_2	0.1375	1.4761	26.425	0.0483
FF3	θ_3	0.1461	1.6623	26.425	0.0483

Table 2: GMM estimation results for CCAPM and FF3 models

Question 5

At the 5% significance level, the CCAPM model is not rejected by the J-test ($p = 0.1012$), while the FF3 model is rejected ($p = 0.0483$). However, even for CCAPM the estimated coefficients are not reliable, because the standard errors are extremely large compared to the point estimates. Hence, non-rejection of the model does not imply precise or economically meaningful parameter estimates.

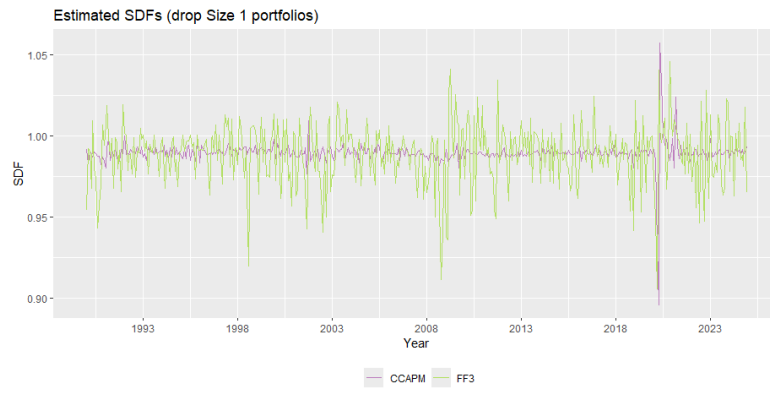


Figure 3: Estimated fitted SDF series for the CCAPM and FF3 models.